GRADE 10 · MATHEMATICS ENGLISH MEDIUM

MARKING SCHEME

Model Paper — 01

Third Term Test (Full Syllabus)

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Note (for teachers):

- Part A: 2 marks for each question (generally 1 mark for the correct method and 1 mark for the correct final answer). — Total 40 marks
- Part B: 10 marks for each question. — Total 60 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. Accept a reasonable range for values read from graphs.

Part A — Answers

1. $\sqrt{2304}$: $23 \rightarrow 4^2 = 16$, remainder 7; $704 \rightarrow 88 \times 8 = 704$, remainder 0

Final answer: 48

(2 marks)

2. $\frac{7}{3} \div \frac{7}{6} \times \frac{3}{4} = \frac{7}{3} \times \frac{6}{7} \times \frac{3}{4} = \frac{3}{2}$

Final answer: $\frac{3}{2} = 1\frac{1}{2}$

(2 marks)

3. $(2x)^2 - 2(2x)(3) + 3^2$

Final answer: $4x^2 - 12x + 9$

(2 marks)

4. $6x^2 - 4x + 3x - 2 = 2x(3x - 2) + 1(3x - 2)$

Final answer: $(3x - 2)(2x + 1)$

(2 marks)

5. L.C.M. of the coefficients = 36 ; a^3, b^3

Final answer: $36a^3b^3$

(2 marks)

6. $\frac{3(x + 1) - 2(x - 2)}{(x - 2)(x + 1)} = \frac{3x + 3 - 2x + 4}{(x - 2)(x + 1)}$

Final answer: $\frac{x + 7}{(x - 2)(x + 1)}$

(2 marks)

7. Arc length = $\frac{120}{360} \times 2 \times \frac{22}{7} \times 21 = 44$ cm ; Perimeter = 44 + 21 + 21

Final answer: 86 cm

(2 marks)

8. $A\hat{B}C = A\hat{C}B = (180° - 40°) \div 2 = 70°$; $A\hat{C}D = 180° - 70°$

Final answer: 110°

(2 marks)

9. $\log_2 2^5 + \log_3 3^{-2} = 5 + (-2)$

Final answer: 3

(2 marks)

10. $x^4 = 81 = 3^4$

Final answer: x = 3

(2 marks)

11. $T_n = a + (n - 1)d$; $T_{15} = 7 + 14 \times 4$

Final answer: 63

(2 marks)

12. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$; $27 = 18 + 15 - n(A \cap B)$

Final answer: 6

(2 marks)

13. Interest = $25\,000 \times \frac{12}{100} \times \frac{5}{2}$

Final answer: Rs. 7 500

(2 marks)

14. Man-days = 12 × 15 = 180 ; 180 ÷ 18

Final answer: 10 days

(2 marks)

15. $2as = v^2 - u^2$

Final answer: a = $\frac{v^2 - u^2}{2s}$

(2 marks)

16. $2x - 3x < 1 + 5 \Rightarrow -x < 6 \Rightarrow x > -6$

Final answer: the least integer value is −5

(2 marks)

17. $\frac{20}{72} \times 360°$

Final answer: 100°

(2 marks)

18. $\frac{4}{10}$

Final answer: $\frac{2}{5}$

(2 marks)

19. Co-interior angles: $(2x + 10) + (3x - 5) = 180 \Rightarrow x = 35$; $\hat{C} = \hat{A} = 2(35) + 10$

Final answer: 80°

(2 marks)

20. $V = \pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 10$

Final answer: 1540 cm^3

(2 marks)

Part B — Answers and Mark Allocation

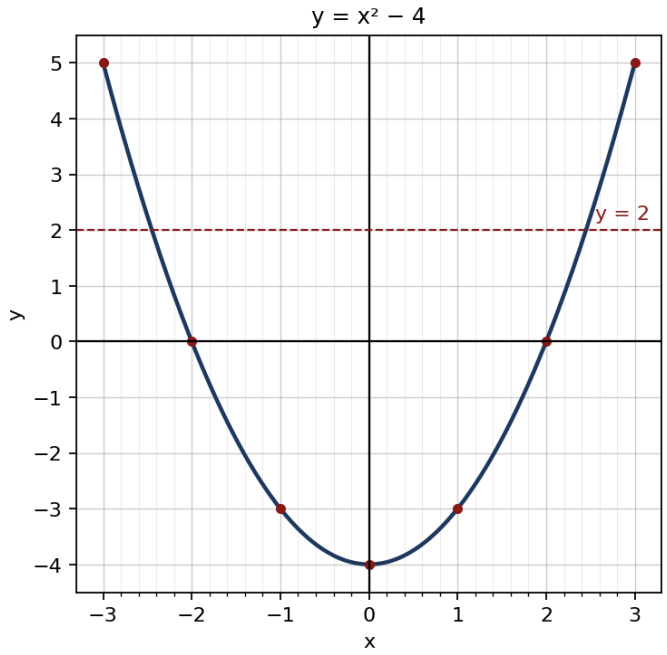
Question 1

(a)	
① × 3: $9x + 6y = 48$; ② × 2: $10x - 6y = 28$	**1**
Adding: $19x = 76$	**1**
$x = 4$	**1**
By substitution: $12 + 2y = 16 \Rightarrow y = 2$	**1**
(b)	
$(x - 7)(x + 2) = 0$	**1**
$x = 7$ or $x = -2$	**1**
(c)	
(i) $x(x + 3) = 54 \Rightarrow x^2 + 3x - 54 = 0$	**1**
(ii) $(x + 9)(x - 6) = 0 \Rightarrow x = 6$ ($x = -9$ is not acceptable)	**1**
Width = 6 m, length = 9 m	**1**
(iii) Perimeter = $2(9 + 6) = 30$ m ; length of wire = $3 \times 30 = 90$ m	**1**

Total marks: 10

Question 2

(a)	
$x = -2 \Rightarrow y = 0$; $x = 1 \Rightarrow y = -3$ (if both are correct)	**1**
(b)	
Correct scale and axes	**1**
Plotting the 7 points correctly	**1**
Smooth curve	**1**



(c)	
(i) Minimum value = −4	**1**
(ii) x = 0	**1**
(iii) $-2 \leq x \leq 2$	**1**
(iv) $x^2 - 6 = 0 \Leftrightarrow x^2 - 4 = 2$; draw the line y = 2 and read: $x \approx \pm 2.45$ (accept ±2.4 – ±2.5)	**1**
$\sqrt{6} \approx 2.45$	**1**
(d)	
$y = x^2 - 1$	**1**

Total marks: 10

Question 3

(a)	
Volume = $2.2 \times 1.4 \times 0.5 = 1.54\ m^3$	**1**
= 1540 litres	**1**
(b)	
Base area = $\frac{22}{7} \times 0.7 \times 0.7 = 1.54\ m^2$	**1**
Height = 1.54 ÷ 1.54	**1**
= 1 m (100 cm)	**1**
(c)	
$2\pi rh = 2 \times \frac{22}{7} \times 0.7 \times 1$	**1**

= 4.4 m^2	**1**
(d)	
Remaining height 0.5 m ; volume = 1.54 × 0.5 = 0.77 m^3	**1**
= 770 litres	**1**
Time = 770 ÷ 35 = 22 minutes	**1**

Total marks: 10

Question 4

(a)	
Diagram with A, B, C, D, O, P, Q correctly included	**1**
(b) In the triangles AOP and COQ,	
OA = OC (the diagonals of a parallelogram bisect each other)	**1**
$A\hat{O}P = C\hat{O}Q$ (vertically opposite angles)	**1**
$O\hat{A}P = O\hat{C}Q$ (AB ∥ DC, alternate angles)	**1**
∴ ΔAOP ≡ ΔCOQ (A.A.S.)	**1**
(c)	
Corresponding sides of congruent triangles ⇒ AP = CQ	**1**
(d)	
AB = DC (opposite sides) ⇒ AB − AP = DC − CQ ⇒ PB = DQ	**1**
Since PB ∥ DQ as well, one pair of opposite sides is equal and parallel ⇒ PBQD is a parallelogram	**1**
(e)	
The diagonals divide the parallelogram into 4 triangles of equal area ; ΔAOB = ¼ × 60	**1**
= 15 cm^2	**1**

Total marks: 10

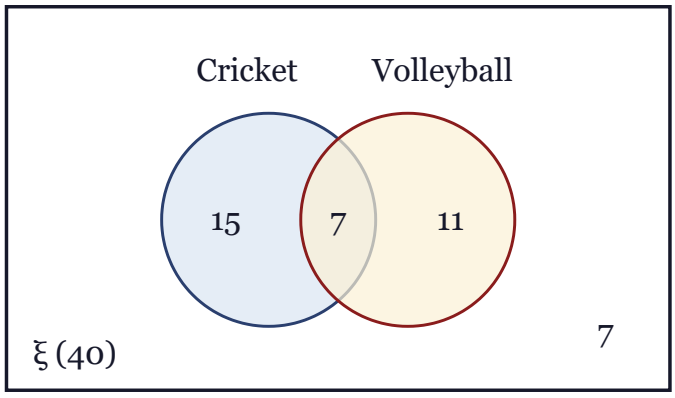
Question 5

(a)	
30 – 40	**1**
(b)	
Class mid-values: 15, 25, 35, 45, 55	**1**
fx: 60, 175, 420, 450, 385	**1**
Σfx = 1490	**1**

Mean = 1490 ÷ 40 = 37.25	**1**
(c)	
40 marks or more = 10 + 7 = 17	**1**
$\frac{17}{40} \times 100 = 42.5\%$	**1**
(d)	
Total marks of the new students = 10 × 47.25 = 472.5	**1**
Total = 1490 + 472.5 = 1962.5	**1**
Mean = 1962.5 ÷ 50 = 39.25	**1**

Total marks: 10

Question 6

(a)	
22 + 18 − x + 7 = 40	**1**
x = 7	**1**
(b)	
	
ξ and the two intersecting circles	**1**
15, 7, 11 correct	**1**
7 outside	**1**
(c)	
22 − 7 = 15	**1**
(d)	
(i) Only one sport = 15 + 11 = 26	**1**
$P = \frac{26}{40} = \frac{13}{20}$	**1**
(ii) Do not play volleyball = 40 − 18 = 22	**1**

$P = \frac{22}{40} = \frac{11}{20}$	**1**

Total marks: 10